

1. SPECTRAL ANALYSIS OF $\theta_j(\lambda)$

In the discrete setting, the distribution of the angles follows the equation

$$2\theta_j(\lambda) - \theta_{j-1}(\lambda) - \theta_{j+1}(\lambda) - (2\theta_j(0) - \theta_{j-1}(0) - \theta_{j+1}(0)) + \lambda \sin \theta_j(\lambda) = 0, \quad (1)$$

where $\theta_j(0) = s_j \alpha$, $s_j = \pm 1$ with equal probability and $\alpha \in (0, \frac{\pi}{2})$.

Since the initial distribution has that $|\theta_j(\lambda)| < |\theta_j(0)| < \frac{\pi}{2}$, we can linearize the system such that

$$2\theta_j(\lambda) - \theta_{j-1}(\lambda) - \theta_{j+1}(\lambda) - (2\theta_j(0) - \theta_{j-1}(0) - \theta_{j+1}(0)) + \lambda \theta_j(\lambda) \approx 0. \quad (2)$$

Next, we introduce the Discrete Fourier Transform (DFT) to solve Eq. (2).

1.1. Discrete Fourier Transform. As a function of j , we assume periodic boundary conditions on the angle sequence, *i.e.* $\theta(\lambda, j + N) = \theta(\lambda, j)$ ¹ is N -periodic on j . As such, we define the N -Discrete Fourier Transform of a N -periodic function $f[j]$ as

$$\hat{f}_N[k] := \mathcal{F}_N\{f\}[k] = \sum_{j=0}^{N-1} f[j] e^{-i\frac{2\pi}{N}jk}. \quad (3)$$

Likewise, it follows that the inverse of the N -DFT is

$$\mathcal{F}_N^{-1}\{\hat{f}_N\}[j] = \frac{1}{N} \sum_{k=0}^{N-1} \hat{f}[k] e^{i\frac{2\pi}{N}kj}. \quad (4)$$

We will often write $\omega_N = \frac{2\pi}{N}$ for notational convenience.

1.2. The power spectrum and correlation function. Making use of the N -DFT on Eq. (2) and using the fact that $\mathcal{F}_N\{f(\cdot - r)\}[k] = e^{-i\omega_N r k} \hat{f}_N[k]$ we see that

$$4 \sin^2(\omega_N k) \left(\hat{\theta}_N(\lambda, k) - \hat{\theta}_N(0, k) \right) + \lambda \hat{\theta}_N(\lambda, k) = 0 \quad (5)$$

$$\Rightarrow \hat{\theta}_N(\lambda, k) = \frac{4 \sin^2(\omega_N k)}{4 \sin^2(\omega_N k) + \lambda} \hat{\theta}_N(0, k). \quad (6)$$

Thus, the spectral power density is

$$\begin{aligned} S_N(\lambda, k) &:= \left| \hat{\theta}_N(\lambda, k) \right|^2 \\ &= \left(\frac{4 \sin^2(\omega_N k)}{4 \sin^2(\omega_N k) + \lambda} \right)^2 \left| \hat{\theta}_N(0, k) \right|^2 \end{aligned} \quad (7)$$

¹We denote that j is a discrete variable with square brackets. As such, the function $\theta(\lambda, j)$ depends on the continuous variable λ and the discrete variable j .

1.2.1. *Power spectrum of $\theta_j(0)$.* It is straightforward to see that the expected value of the initial distribution is

$$\mathbb{E}[\theta_j(0)] = P(\theta_j(0) = \alpha) \cdot (\alpha) + P(\theta_j(0) = -\alpha) \cdot (-\alpha) = 0. \quad (8)$$

Likewise, its second moment is

$$\mathbb{E}[(\theta_j(0))^2] = P(\theta_j(0) = \alpha) \cdot (\alpha)^2 + P(\theta_j(0) = -\alpha) \cdot (-\alpha)^2 = \alpha^2. \quad (9)$$

The initial correlation function is then

$$R(0, r] = \mathbb{E}[\theta_j(0)\theta_{j-r}(0)] = \alpha^2\delta_{r,0}, \quad (10)$$

and consequently, its spectral power density is

$$\hat{R}_N(0, k] = \alpha^2 = \mathbb{E}[S_N(0, k)]. \quad (11)$$

1.2.2. *Power spectrum of $\theta_j(\lambda)$.* The correlation function of $\theta_j(\lambda)$ is defined as

$$R(\lambda, r] = \mathbb{E}[\theta_j(\lambda)\theta_{j-r}(\lambda)]. \quad (12)$$

Taking the N -DFT, and using Eqs. (8) and (11) we see that

$$\begin{aligned} \hat{R}_N(\lambda, k] &= \mathbb{E}\left[\left(\frac{4\sin^2(\omega_N k)}{4\sin^2(\omega_N k) + \lambda}\right)^2 \left|\hat{\theta}^N(0, k]\right|^2\right] \\ &= \left(\frac{4\sin^2(\omega_N k)}{4\sin^2(\omega_N k) + \lambda}\right)^2 \mathbb{E}[S_N(0, k)] \\ &= \left(\frac{4\sin^2(\omega_N k)}{4\sin^2(\omega_N k) + \lambda}\right)^2 \hat{R}_N(0, k] \\ &= \alpha^2 \left(\frac{4\sin^2(\omega_N k)}{4\sin^2(\omega_N k) + \lambda}\right)^2 \end{aligned} \quad (13)$$

$$= \mathbb{E}[S_N(\lambda, k)]. \quad (14)$$

We are now ready to proceed with the scaling of the deformation.

2. DEFORMATION SCALING

At fixed tension $\lambda \in [0, \infty)$, we have that the x -axis length of the chain is

$$L(\lambda) = \sum_{j=0}^{N-1} \ell \cos(\theta_j(\lambda)). \quad (15)$$

When $\lambda = 0$, we have that

$$L(0) = \sum_{j=0}^{N-1} \ell \cos(\theta_j(0)) = \sum_{j=0}^{N-1} \ell \cos(s_j \alpha) = N \ell \cos(\alpha). \quad (16)$$

In terms of the initial length, we write

$$L(\lambda) = (1 + \epsilon(\lambda))L(0). \quad (17)$$

Therefore, the deformation $\epsilon(\lambda)$ is

$$\epsilon(\lambda) = \frac{L(\lambda) - L(0)}{L(0)}. \quad (18)$$

Recalling that $|\theta_j(\lambda)| < \frac{\pi}{2}$, we estimate $\cos \theta_j(\lambda) \approx 1 - \frac{\theta_j(\lambda)^2}{2}$. Then, we have that

$$\epsilon(\lambda) \approx \frac{1}{2N \cos(\alpha)} \sum_{j=0}^{N-1} (\theta_j(0)^2 - \theta_j(\lambda)^2) \quad (19)$$

2.1. Error estimation. Heuristically, we will estimate the real-space average of a function $\frac{1}{N} \sum_{j=0}^{N-1} f[j]$ with the statistical average of the sequence of measurements $\mathbb{E}[f_j]$. Formally,

$$\frac{1}{N} \sum_{j=0}^{N-1} f[j] \approx \mathbb{E}[f_j]. \quad (20)$$

Quantitatively, we seek the error on this estimation to be a fraction of the fluctuations of f_j . If the Central Limit Theorem applies, the standard error of the mean is given by

$$\eta_f[N] = \frac{\sigma_f}{\sqrt{N}}, \quad (21)$$

where

$$\sigma_f := \sqrt{\mathbb{E}[(f_j)^2]}$$

Then, writing the error in units of σ_f at a fraction $\mu[N] \in (0, 1]$, we have that

$$\eta[N] = \mu[N] \sigma_f \Rightarrow \mu[N] = N^{-\frac{1}{2}}. \quad (22)$$

In the case of $\sigma_\theta(\lambda) = \mathcal{O}(\alpha)$ and thus all N scaling is concentrated on $\mu[N]$. Then, up to an error of $\mu[N] = \mathcal{O}(N^{-1/2})$, we have that ²

$$\begin{aligned}\epsilon(\lambda) &\approx \frac{1}{2\cos(\alpha)} \frac{1}{N} \sum_{j=0}^{N-1} (\theta_j(0)^2 - \theta_j(\lambda)^2) \\ &= \frac{1}{2\cos(\alpha)} \mathbb{E} [\theta_j(0)^2 - \theta_j(\lambda)^2] + \mathcal{O}(N^{-\frac{1}{2}}).\end{aligned}\quad (23)$$

2.2. Deformation formula. Our purpose is to determine the scaling of the deformation $\epsilon(\lambda)$ in terms of λ in the regime $\lambda \leq 1$.

Using Eq. (12), we find that the second moment of $\theta_j(\lambda)$ is

$$\mathbb{E} [\theta_j(\lambda)^2] = R(\lambda, 0). \quad (24)$$

As a Fourier series, we write

$$R(\lambda, r) = \frac{1}{N} \sum_{k=0}^{N-1} \hat{R}_N(\lambda, k) e^{i\omega_N k r}. \quad (25)$$

As such, we have the zero-mode value

$$R(\lambda, 0) = \frac{\omega_N}{2\pi} \sum_{k=0}^{N-1} \hat{R}_N(\lambda, k). \quad (26)$$

Using the above, we write the deformation as

$$\epsilon(\lambda) \approx \frac{1}{4\pi \cos(\alpha)} \sum_{k=0}^{N-1} (\hat{R}_N(0, k) - \hat{R}_N(\lambda, k)) \quad (27)$$

$$= \frac{\alpha^2}{4\pi \cos(\alpha)} \sum_{k=0}^{N-1} \left(1 - \left(\frac{4 \sin^2(\omega_N k)}{4 \sin^2(\omega_N k) + \lambda} \right)^2 \right) \omega_N. \quad (28)$$

2.3. Quadratic relation. Numerics suggests that the quadratic relation is mostly independent of N , therefore we will take the thermodynamic limit of our expression. As $N \rightarrow \infty$, $\omega_N \rightarrow 0$ and we have that the sum converges to the Riemann integral

$$\epsilon(\lambda) \approx \frac{\alpha^2}{4\pi \cos(\alpha)} \int_0^{2\pi} \left(1 - \left(\frac{\sin^2(\xi)}{\sin^2(\xi) + \frac{\lambda}{4}} \right)^2 \right) d\xi \quad (29)$$

$$= \sqrt{\lambda} \frac{\alpha^2}{2\cos(\alpha)} \frac{6 + \lambda}{(4 + \lambda)^{3/2}} \quad (30)$$

$$\approx \frac{\alpha^2}{2\cos(\alpha)} \left(\frac{3}{4} \sqrt{\lambda} + \mathcal{O}\left(\lambda^{\frac{3}{2}}\right) \right), \quad (31)$$

²En este caso, podemos tomar $\mathbb{E}[\epsilon(\lambda)]$ y sería lo mismo por que la cantidad es "self-averaging".

where we expanded around $\lambda \sim 0$ on the last step. Finally, we can invert the relation which yields the tension-deformation relation

$$\lambda(\epsilon) \approx \frac{4 \cos^2(\alpha)}{\alpha^4} \left(\frac{16}{9} \epsilon^2 + \mathcal{O}(\epsilon^4) \right). \quad (32)$$

Define $A(\alpha) := \left(\frac{8 \cos(\alpha)}{3 \alpha^2} \right)^2$. The quadratic relation is shown in red in Figure 1.

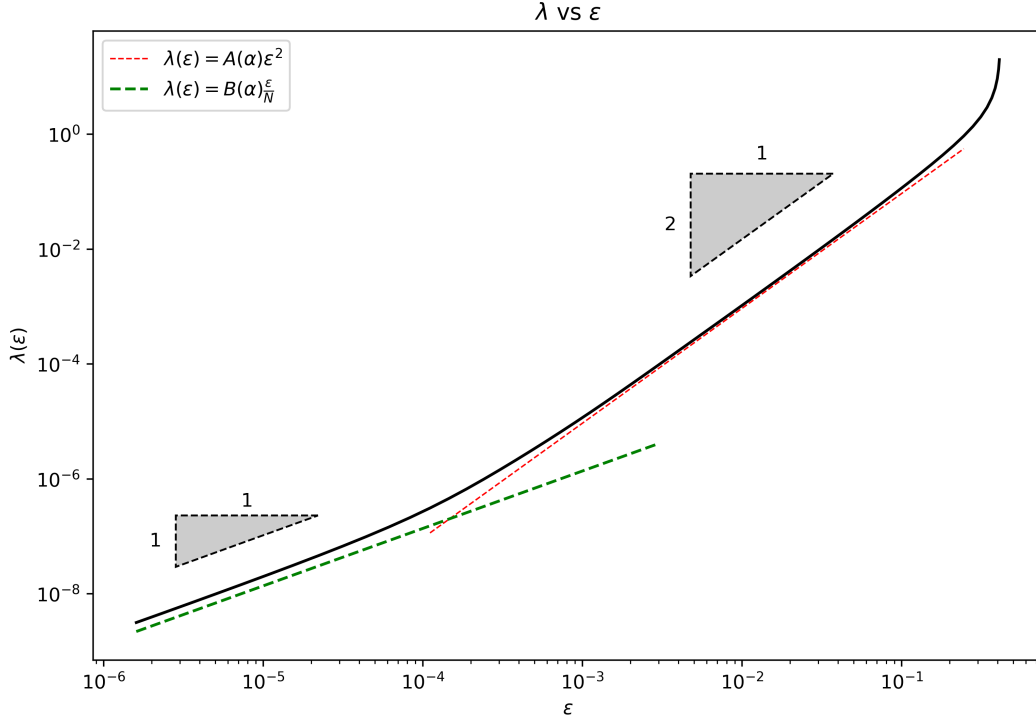


FIGURE 1. (black) Deformation vs tension curve for $N = 10000$. (red) Analytical estimation of the quadratic tension relation $\lambda(\epsilon) \approx A(\alpha)\epsilon^2$. (green) Analytical estimation of the linear tension relation $\lambda(\epsilon) \approx B(\alpha)\frac{\epsilon}{N}$.

2.4. Linear relation. Starting from Eq. (28), we want to probe the deformation behavior for small force λ . We will first single out the points $k = 0, N/2$ in the expression and then expand the sum around $\lambda \sim 0$. We write

$$\epsilon(\lambda) \approx \frac{\alpha^2 \omega_N}{4\pi \cos(\alpha)} \left(2 + \sum_{k=1}^{N/2-1} \left[1 - \left(\frac{4 \sin^2(\omega_N k)}{4 \sin^2(\omega_N k) + \lambda} \right)^2 \right] + \sum_{k=N/2}^{N-1} \left[1 - \left(\frac{4 \sin^2(\omega_N k)}{4 \sin^2(\omega_N k) + \lambda} \right)^2 \right] \right). \quad (33)$$

Considering $\lambda \ll 1$, we expand our functions as

$$\epsilon(\lambda) \approx \frac{\alpha^2 \omega_N}{4\pi \cos(\alpha)} \left(2 + \lambda \sum_{k=0}^{N/2-1} \csc^2(\omega_N k) + \lambda \sum_{k=N/2}^{N-1} \csc^2(\omega_N k) \right). \quad (34)$$

Using the π -periodicity of $\csc(x)$, and the fact that

$$\sum_{k=1}^{M-1} \csc^2\left(\frac{\pi}{M}k\right) = \frac{M^2 - 1}{3}, \quad (35)$$

we have that

$$\epsilon(\lambda) \approx \frac{1}{N} \frac{\alpha^2}{\cos(\alpha)} \left[1 + \frac{\lambda}{3} \left(\frac{N^2}{4} - 1 \right) \right]. \quad (36)$$

Inverting the relation and given in terms of orders of N ,

$$\lambda(\epsilon) \approx \frac{12 \cos(\alpha)}{\alpha^2 N} \epsilon + \mathcal{O}(N^{-2}). \quad (37)$$

Define $B(\alpha) = \frac{12 \cos(\alpha)}{\alpha^2}$. Figure 1 shows the linear tension-deformation relation in green.